

WEEK-2 TASK

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Student Details

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Aim

To develop a Dynamic Contextual Pricing Engine using the Thompson Sampling algorithm that selects the best product price based on customer purchase behavior and market conditions.

Objective

- To simulate dynamic pricing.

- To maximize customer conversions.
- To apply the Thompson Sampling algorithm.
- To balance exploration and exploitation.
- To automatically adjust prices according to competitor actions.

Introduction

Dynamic pricing is a pricing strategy where the price of a product changes based on customer demand, buying behavior, and market conditions. It helps businesses increase sales and maximize profits.

In this project, different product prices are treated as different arms of a Multi-Armed Bandit. The Thompson Sampling algorithm learns from customer purchase behavior and chooses the best price. If a competitor changes its price, the algorithm updates the product price automatically.

Problem Statement

Many businesses use fixed prices, which may reduce sales and profit. A dynamic pricing system can continuously learn from customer responses and select the best price to improve revenue and customer satisfaction.

Requirements

Functional Requirements

- Initialize multiple product prices.
- Collect customer purchase data.
- Calculate successful and unsuccessful purchases.
- Apply the Thompson Sampling algorithm.
- Select the best price.
- Update prices based on competitor changes.

Non-Functional Requirements

- Fast decision making.
- High accuracy.
- Easy to use.
- Reliable performance.
- Scalable for different products.

Algorithm

Step 1: Initialize different product prices.

Step 2: Observe customer purchase behavior.

Step 3: Count successful purchases and unsuccessful purchases.

Step 4: Update the Beta Distribution.

Step 5: Apply Thompson Sampling.**Step 6:** Select the product price with the highest probability.**Step 7:** Monitor competitor pricing.

Step 8: Update the selected price if necessary.**Step 9:** Repeat the process continuously.

Start



Initialize Product Prices



Collect Customer Data



Calculate Success & Failure



Update Beta Distribution



Apply Thompson Sampling



Select Best Price



Competitor Changes Price?



Yes / No



Update Product Price



End

Price (₹)	Custom ers	Bou ght	Not Bought
100	20	18	2
120	20	15	5
140	20	10	10
160	20	5	15

```
import random
```

```
prices = [100, 120, 140, 160]
```

```
success = [18, 15, 10, 5]
```

```
failure = [2, 5, 10, 15]
```

```
best_price = None
```

```
best_value = 0
```

```
for i in range(len(prices)):
```

```
    sample = random.betavariate(success[i] + 1, failure[i] + 1)
```

```
    if sample > best_value:
```

```
        best_value = sample
```

```
        best_price = prices[i]
```

```
print("Selected Price:", best_price)
```

Sample Output

Selected Price : 100

Note: The output may change every time because Thompson Sampling uses random sampling.

Competitive Shock Example

Initially, the company sells a product at ₹120 while the competitor sells it at ₹130.

Later, the competitor reduces the price to ₹100.

The Thompson Sampling algorithm analyzes customer responses and selects a new competitive price such as ₹110 to maintain sales and profit.

Advantages

- Improves pricing decisions.
- Maximizes profit.
- Increases customer satisfaction.
- Learns automatically from customer behavior.
- Responds quickly to market changes.
- Reduces manual pricing effort.

Applications

- Amazon
- Flipkart
- Swiggy
- Zomato
- Uber
- Airline ticket booking
- Hotel booking websites
- Online retail stores

Conclusion

The Dynamic Contextual Pricing Engine using Thompson Sampling is an intelligent pricing system that learns from customer behavior and market conditions. It helps businesses select the best product price, improve customer satisfaction, increase sales, and maximize profits. This approach provides a smart and efficient solution for modern business pricing strategies.